



FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

**ECONOMICS 114 A1**

|                    |                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| DATE:              | 11 April 2023                                                                                                |
| DURATION:          | 2 hours                                                                                                      |
| TOTAL OF MARKS:    | 60                                                                                                           |
| EXAMINERS:         | Prof S du Plessis (convenor)<br>Ms L Boltman<br>Dr C Links<br>Ms W Nkosi<br>Dr D Shepherd<br>Mr M van Schoor |
| INTERNAL EXAMINER: | Prof P de Villiers                                                                                           |

**PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.**

ANSWER **SECTION A** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO, ANSWER **SECTION B** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS NEXT** TO THE NUMBER OF THE STATEMENT.

ANSWER **SECTION C** QUESTIONS IN THE ANSWER BOOK PROVIDED. EVERY QUESTION SHOULD START ON A NEW PAGE. SHOW ALL CALCULATIONS

**KEY:**

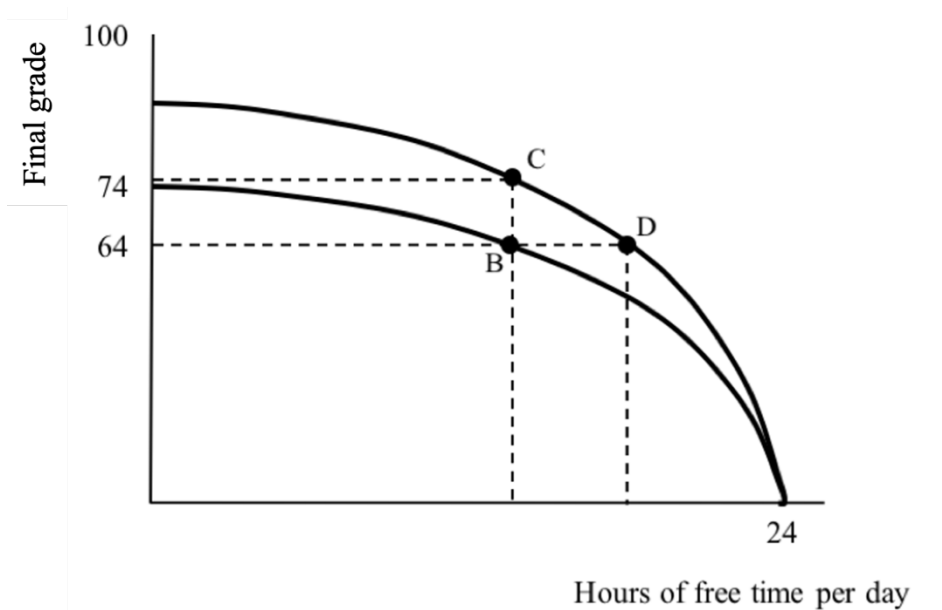
AP = Average product  
BL = Budget line  
GDP = Gross domestic product  
IC = indifference curve  
K = capital  
L = labour  
FF = Feasible frontier  
MP = Marginal product

MRS = Marginal rate of substitution  
MRT = Marginal rate of transformation  
P = Price  
PF = Production function  
Q = Quantity  
TP = Total product  
w = Wage  
Y = Income

**SECTION A**  
**TRUE/FALSE STATEMENTS**

**[6 marks: 1 mark for a correct answer and - ½ for an incorrect answer]**

1. The Industrial Revolution first occurred in Britain, and then sometime later in Japan.  
a) True  
b) False
  
2. The Malthusian trap is an economic theory that suggests, that in the long run, an increase in productivity will result in higher wages and a subsequent rise in living standards.  
a) True  
b) False
  
3. Carlyssa learns a new study technique that allows her to achieve a higher mark for every given number of hours she spends studying. This is shown by the outward shift of her feasible frontier in the figure below.



Based on this information, we can say with certainty that Carlyssa will derive more utility from option C than she would from option D.

- a) True
- b) False

4. The MRS indicates the change in the consumer's utility when one good is substituted for another.
- a) True
  - b) False

5. Consider the following game table that describes an interaction between two housemates, Anthony Patch and Gloria Gilbert:

|               |                    | Gloria Gilbert     |             |
|---------------|--------------------|--------------------|-------------|
|               |                    | Shop for groceries | Cook dinner |
| Anthony Patch | Shop for groceries | 2, 2               | 4, 3        |
|               | Cook dinner        | 1, 0               | 1, 1        |

Anthony Patch has a dominant strategy of Shop for groceries.

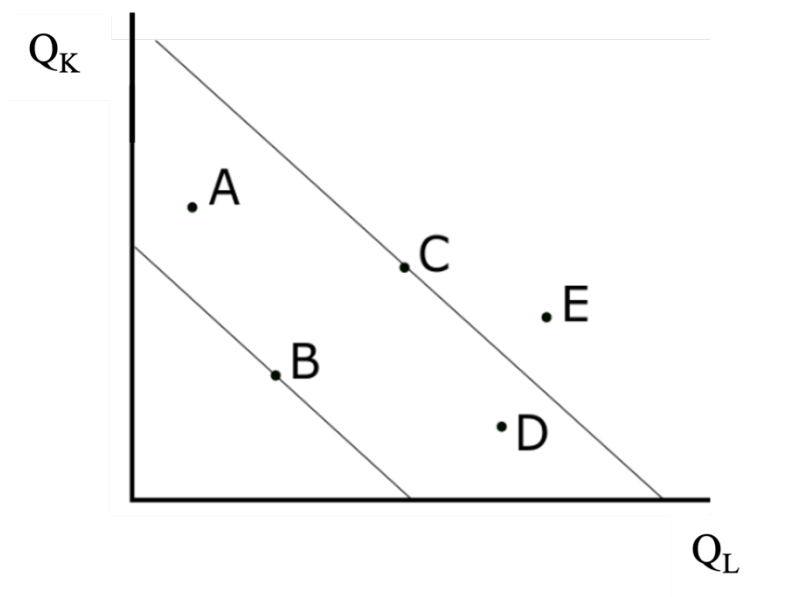
- a) True
  - b) False
6. The veil of ignorance enables you, in making a judgment about fairness, to evaluate the constitutions, laws, inheritance practices, and other institutions of society as an impartial outsider.
- a) True
  - b) False

## SECTION B MULTIPLE CHOICE QUESTIONS

**[24 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]**

7. Ukraine's GDP in 2021 was \$200 billion. In 2022 it is estimated that Ukraine's GDP had declined by 35% due to the ongoing war. Ukraine's population was estimated at 41.4 million in 2021, but it is believed that approximately 8 million people have left Ukraine between 2021 and 2022. Which ONE of the statements below is FALSE?
- a) Population decreased by approximately 19.3%.
  - b) GDP per capita increased by approximately 19.4%.
  - c) The percentage decrease in population is smaller than the percentage decrease in the GDP.
  - d) GDP per capita in 2022 is approximately \$3 892.

8. eSwatini and South Africa both produce sugar and beef. If eSwatini devoted all its resources to producing sugar, it could produce one million tonnes per year, while if it devoted all its resources to producing beef, it could produce 7 000 tonnes per year. South Africa could produce 12 million tonnes of sugar per year or 220 000 tonnes of beef per year if using all its resources to produce only one product. The opportunity cost of a tonne of beef in the country that has a comparative advantage in the production of beef is
- 142 tonnes of sugar
  - 0.02 tonnes of beef
  - 0.000006 tonnes of beef
  - 54.5 tonnes of sugar
9. The following is a definition of \_\_\_\_\_:
- “A model outcome that is self-perpetuating and does not change unless an outside or external force is introduced that alters the model’s description of the situation”.
- An equilibrium
  - Subsistence level
  - A reservation option
  - None of the above
10. The graph below shows the five different available technologies for producing a thousand litres of soft drink per day.  $Q_L$  measures the number of workers and  $Q_K$  measures the number of bottling machines used. In addition, two isocost curves, reflecting the current price of bottling machines and the wage rate, are shown.

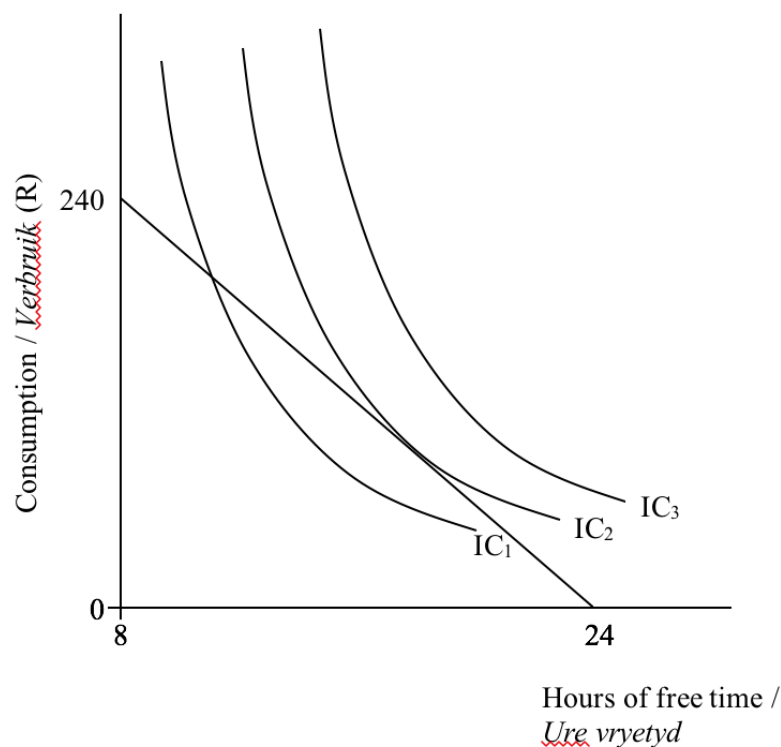


Which of the following statements are TRUE?

- i) **D** is the cheapest available technology.
- ii) A firm would make more profit using technology **A** than using technology **E**.
- iii) If a single firm had a patent right on technology **B**, so that it was the only firm able to use technology **B**, that firm would earn an innovation rent.
- iv) Technology **D** could never become the cheapest available technology, even if the relative prices of the inputs do change.

- a) only ii
- b) iv and i
- c) ii and iii
- d) only iv

11. During the end-of-year holidays, you find work selling ice creams for a shop next to the beach. The company tells you that you will be paid per hour and that your hours are flexible; that is, you can choose how much you want to work. The figure below illustrates your budget constraint and several indifference curves.



What is the hourly wage rate that the company will pay you?

- a) R34
- b) R27
- c) R15
- d) R30

12. Alana chooses how many hours to spend studying for a test. If the final grade she can achieve is given by a concave production function, this implies that the marginal product of each hour she uses to study \_\_\_\_\_ as the number of hours of study \_\_\_\_\_.

- a) increases, increase
- b) increases, decrease
- c) decreases, decrease
- d) decreases, increase

13. A consumer who has a fixed income in Rands, consumes coffee and bread. Suppose that coffee is a normal good, but bread is an inferior good. When the price of coffee increases, *ceteris paribus*, the substitution effect on the consumption of bread is \_\_\_\_\_, while the income effect on bread is \_\_\_\_\_.

- a) positive, negative
- b) negative, positive
- c) negative, negative
- d) positive, positive

14. The table below represents Palesa's feasible frontier for the production of soft drinks.

|                        |     |     |     |     |     |
|------------------------|-----|-----|-----|-----|-----|
| Hours worked           | 20  | 24  | 28  | 30  | 32  |
| Bottles of soft drinks | 180 | 300 | 380 | 460 | 500 |

The marginal product of the 30<sup>th</sup> hour worked is approximately \_\_\_\_ soft drinks.

- a) 80
- b) 20
- c) 30
- d) 460

15. T'Challa and M'Baku are candidates for the presidency of Wakanda in 2023, and they must decide what type of campaign they are going to run. They can choose to run a constructive campaign where they promote their own policy stances, or they can choose to run a slanderous campaign where they put down their opponent by attacking their character in the media. The strategies and associated payoffs are represented in the table below.

|          |              | M'Baku       |            |
|----------|--------------|--------------|------------|
|          |              | Constructive | Slanderous |
| T'Challa | Constructive | 2, 6         | 2, 1       |
|          | Slanderous   | 3, 4         | 5, 1       |

The Nash Equilibrium of this game is

- a) (Constructive, Constructive)
- b) (Slanderous, Constructive)
- c) (Constructive, Slanderous)
- d) (Slanderous, Slanderous)

16. Master KG and Lady Zamar are performers who have decided to collaborate on a new song. To do so, they must decide which part of the new song they want to write, as this will determine which part of the song they will have to perform. Their decision is represented by the game table below.

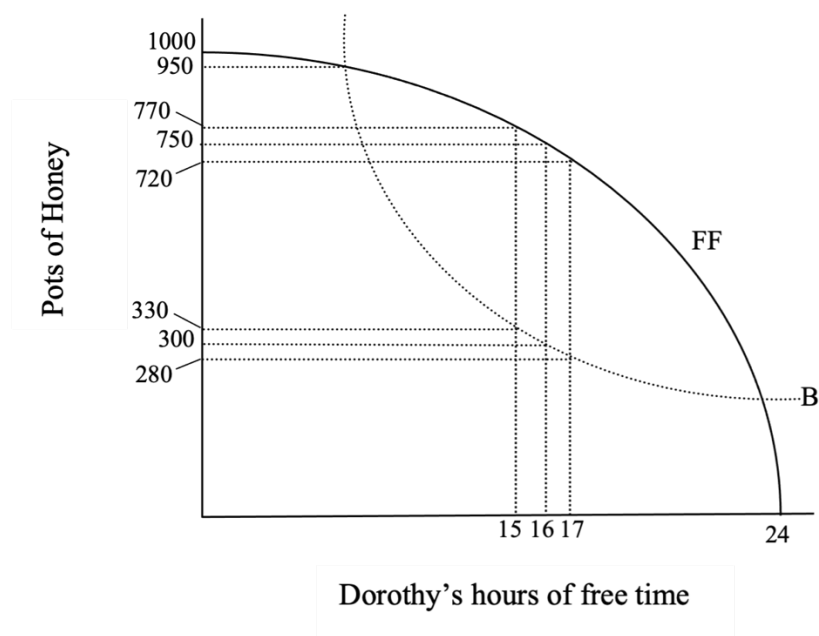
|           |        | Lady Zamar |       |
|-----------|--------|------------|-------|
|           |        | Chorus     | Verse |
| Master KG | Chorus | 3, 2       | 4, 5  |
|           | Verse  | 2, 2       | 3, 3  |

Which of the following statements regarding the game above are TRUE?

- i. Master KG has a dominant strategy to write the chorus.
- ii. Lady Zamar has a dominant strategy to write the chorus.
- iii. The Nash Equilibrium of this game is also a dominant strategy equilibrium.
- iv. In Nash Equilibrium, Master KG receives a payoff of 5.
- v. This game is an example of a prisoner's dilemma.

- a) i, ii, iii and v
- b) i, iii, iv and v
- c) i, ii and v
- d) i and iii

17. The Wicked Witch of the West forces Dorothy to farm honey in the Land of Oz. The Witch can choose the number of hours per week that Dorothy must work, and she can choose how many jars of honey Dorothy can take for herself. The graph below indicates the feasible frontier (FF) and Dorothy's biological survival constraint (B).



What is the maximum number of honey jars that the Wicked Witch of the West can get from this arrangement?

- a) 950
- b) 770
- c) 750
- d) 450

18. Chance, Kendrick, and Drake discuss how to share three apples and three oranges. Which ONE of the following statements regarding Pareto optimal allocations is CORRECT?

- a) If Chance likes both apples and oranges, there is no Pareto improvement on an allocation where he has all the apples and all the oranges.
- b) It is always Pareto efficient for Chance, Kendrick, and Drake to have one apple and one orange each.
- c) If at least one of them likes both apples and oranges, there is no Pareto-efficient allocation.
- d) There cannot be more than one Pareto-efficient allocation.

### SECTION C LONG QUESTIONS (30 marks)

#### Question 1

Name TWO different ways that an economist can see that technological progress has taken place when the production function has shifted. [2]

#### Question 2

A farmer produces 10 crates of apples using two inputs, labour (L) and fertiliser (F). The wage rate is R80 per hour and the price of fertilizer is R200 per bag. The farmer has access to three technologies:

|              | Bags of fertiliser | Hours of labour |
|--------------|--------------------|-----------------|
| Technology A | 2                  | 5               |
| Technology B | 3                  | 6               |
| Technology C | 1                  | 7               |

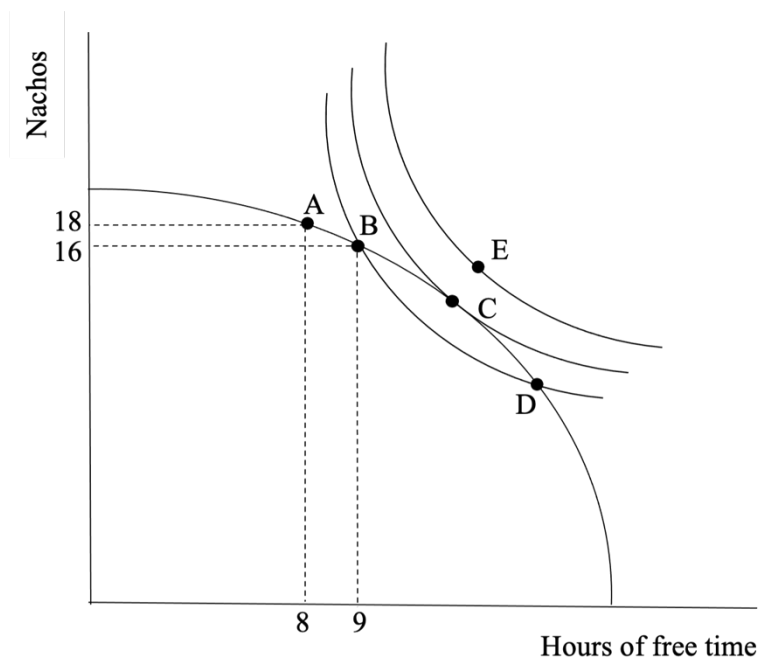
- 2.1 Which technologies are dominated and why? [1]
- 2.2 If fertiliser is plotted on the vertical (y-) axis, derive the equation of the isocost line for the farmer with a total cost of R36 000. [2]

Assume that the farmer is currently utilising technology C. The wage rate now increases from R80 to R120 per hour.

- 2.3 How does this wage increase affect the relative price of labour? [1]
- 2.4 Determine which technology the farmer will employ after the wage increase. [3]
- 2.5 What is the innovation rent associated with adopting this technology? [1]

### Question 3

The graph below indicates the feasible frontier of James, who produces and consumes nachos, as well as his indifference curves.



- 3.1 Which point on the graph above is James's most preferred point? Explain your answer. [2]
- 3.2 What is James's opportunity cost of an hour of free time at point A? [1]
- 3.3 What is James's optimal decision point? [1]

### Question 4

Alissa receives an income of R196 which she can spend on popcorn and lollipops. The price of a bag of popcorn is R14 and the price of a lollipop is R4. Assume that Alissa spends her full income on these two products.

- 4.1 Write down an equation for Alissa's budget line if popcorn is shown on the x-axis. [2]
- 4.2 What does the slope of her budget line indicate? [1]

- 4.3 If Alissa's optimal consumption point is 6 bags of popcorn and 28 lollipops, what is her MRS at this optimal point? [1]
- 4.4 If the price of popcorn decreases to R7 a bag, explain why Alissa would then be able to consume more of both goods. [2]

### Question 5

Consider the following payoff matrix, which depicts a strategic interaction between two friends who are trying to decide where they should buy lunch.

|         |              | Zama         |             |
|---------|--------------|--------------|-------------|
|         |              | Chick Fillet | Texas Grill |
| Ronelle | Chick Fillet | 5, 5         | 3, 2        |
|         | Texas Grill  | 6, 3         | 4, 4        |

- 5.1 Solve for the Nash Equilibrium of this game. [2]
- 5.2 Does the game represent a social dilemma? Explain your answer. [3]

### Question 6

Oliver works for Trixie on her farm producing tulips. He pays Trixie part of the harvest as part of a sharecropping rent agreement. The maximum feasible production and Oliver's preferences are represented in the figure below.  $I_1$  is Oliver's reservation indifference curve (RIC). At every point on the line between **D** and **E**,  $MRS = MRT$ . Explain why this means that these points are Pareto efficient by showing that Pareto improvements from them are not possible. [5]

